

Chapter 5

Mathematical Expectation

Learning Objectives

- Understand and compute the expected values of random variables using engineering data.

5.0 Mathematical Expectation

5.1 Introduction

A very important concept in probability and statistics is that of mathematical expectation, expected value or briefly expectation of a random variable. The expectation of X is very often called the mean of X and is denoted by μ_x or simply μ when a particular random variable is understood. This expected value of x gives a simple value, which acts as a representative, or average of the value of x and for this reason it is often called a measure of central tendency.

Consider that the random variable x has the values x_1, x_2, x_n and $f(x_1), f(x_2), \dots, f(x_n)$ as the probabilities of occurrence. The mean or expected value of x is:

$$\mu = E(X) = \sum x f(x) \quad \text{if } X \text{ is discrete}$$

and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \text{if } X \text{ is continuous}$$

It is said that the expectation $E(X)$ exist for a continuous distribution if and only if the integral is absolutely convergent, that is if and only if $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$

Example 54 (Spiegel, 1980)

Suppose that a game is played with a single die assumed fair. In this game a player wins \$20 if a 2 turns up, \$40 if a 4 turns up; loses \$30 if a 6 turns up; while he neither wins nor loses if any other face turns up. Find the expected sum of money won.

Solution

Table 5.1

I	1	2	3	4	5	6
X_i	0	20	0	40	0	-30
$f(x_i)$	1/6	1/6	1/6	1/6	1/6	1/6

The probability function of x is displayed Table 7.1 Thus the expected value is:

$$\sum x_i f(x_i) = (0) \left(\frac{1}{6} \right) + (20) \left(\frac{1}{6} \right) + (0) \left(\frac{1}{6} \right) + (40) \left(\frac{1}{6} \right) + (-30) \left(\frac{1}{6} \right) = 5$$

It follows that the player can expect to win 5. In a fair game, therefore he should be expected to pay \$5 in order to play the game.

Example 55

Suppose that the pdf of a random variable X with a continuous contribution is $f(x)=$

$$\begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Then

$E(X)=$

$$\int_0^1 xf(x) dx = \int_0^1 x(2x) dx = \left. \frac{2}{3}x^3 \right|_0^1 = \frac{2}{3}$$

5.2 The Expectation of a Function

Let us consider a new random variable $g(x)$, which depends on X , that is, each value of $g(x)$ is determined by knowing the values of x . For instance, $g(x)$ might be x^2 or $3x - 1$, so that whenever X assumes the value 2, $g(X)$ assumes the value $g(2)$. In particular, if X is a discrete random variable with probability distribution $f(x)$, $x=-1, 0, 1, 2$ and $g(X) = X^2$ then

$$P[g(X) = 0] = P(x = 0) = f(0)$$

$$P[g(X) = 1] = p(x = -1) + p(x = 1) = f(-1) + f(1)$$

$$P[g(X) = 4] = p(x = 2) = f(2)$$

So that the probability distribution of $g(x)$ may be written

$g(x)$	0	1	4
$P[g(X)=g(x)]$	$f(0)$	$f(-1)+f(1)$	$f(2)$

By definition of an expected value of a random variable, we obtain

$$\mu_{g(x)} = E[g(x)]$$

$$g(x) = 0f(0) + 1[f(-1) + f(1)] + 4f(2)$$

$$= (-1)^2f(-1) + (0)^2f(0) + (1)^2f(1) + (2)^2f(2)$$

$$= \sum_x g(x)f(x)$$

In the general form, for a random variable X with a probability distribution $f(x)$, the expectation of the random variable $g(X)$ is:

$$\mu_{g(x)} = E[g(X)] = \sum g(x)f(x)$$

If X is discrete, and

$$\mu_{g(x)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

If X is continuous.

Example 56

Suppose that the pdf of a random variable X with a continuous distribution is

$$f(X) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Determine the expectations of Y when $Y = X^{1/2}$

Solution

$$E[Y] = \int_{x=0}^1 g(x)f(x) dx = \int_{x=0}^1 x^{\frac{1}{2}}(2x) dx = \frac{4}{5}x^{\frac{5}{2}} \Big|_0^1 = \frac{4}{5}$$

Example 57

Suppose that the number of cars, X_1 that pass through a car wash between 4pm and 5pm on any sunny Friday has the following probability distribution:

X	4	5	6	7	8	9
f(x)	1/12	1/12	1/4	1/4	1/6	1/6

Let $g(x) = 2X-1$ represents the amount of money in the dollars paid to the attendants expected earnings for the particular time period. How much does the attendant expect to earn on any sunny Friday?

Solution

The attendant can expect to receive

$$\begin{aligned} E[g(X)] &= E(2x - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\ &= 7 \left(\frac{1}{12} \right) + 9 \left(\frac{1}{12} \right) + 11 \left(\frac{1}{4} \right) + 13 \left(\frac{1}{4} \right) + 15 \left(\frac{1}{6} \right) + 17 \left(\frac{1}{6} \right) \\ &= \$12.67 \end{aligned}$$

Example 58

Let X be a continuous random variable with density functions f

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of $g(X) = 4x + 3$

Solution

$$\begin{aligned} E[g(x)] &= \int g(x)f(x) dx \\ E(4x + 3) &= \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx \\ &= \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx \end{aligned}$$

$$\begin{aligned}
&= \left. \frac{x^4 + x^3}{3} \right|_{-1}^2 \\
&= \frac{24}{3} = 8
\end{aligned}$$

5.3 Properties of Expectations

Basic Theorems

Suppose that X is a random variable for which the expectation $E(X)$ exists. We shall present several results pertaining to the basic properties of expectations.

Theorem 1

If $Y = ax + b$ where a and b are constants, then $E(Y) = aE(x) + b$.

Proof: First we shall assume, for convenience, that X has a continuous distribution for which the pdf is $f(x)$. Then,

$$E(Y) = E(ax + b) =$$

$$\begin{aligned}
&\int_{-\infty}^{\infty} (ax + b)f(x) dx \\
&= a \int_{-\infty}^{\infty} xf(x) dx + b \int_{-\infty}^{\infty} f(x) dx \\
&= aE(x) + b
\end{aligned}$$

A similar proof can be given for a discrete distribution or for a more general type of distributions

Example 59

Suppose that $E(x) = 5$ then

$$(i) E(3x - 5) = 3E(X) - 5 = 3(5) - 5 = 10$$

$$(ii) E(-3x + 15) = -3E(x) + 15 = -3(5) + 15 = 0$$

Theorem 2

If there exists a constant a such that $P(x \geq a) = 1$, then $E(x) \geq a$. If there exists a constant b such that $P(x \leq b) = 1$, then

$$E(X) \leq b$$

Theorem 3

If $X_1, X_2, \dots, X_n$ are n random variables such that each expectation $E(X_i)$ exists ($i = 1, 2, \dots, n$), then

$$E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n)$$

This theorem means that the expectations of the sum of several random variables must be equal to the sum of their individual expectations, regardless of whether or not the random variables are independent.

Follow Theorems 1 and 3:

$$E(a_1x_1 + \dots + a_nx_n + b) = a_1E(x_1) + \dots + a_nE(X_n) + b$$

Theorem 4

E(

$$E\left(\prod_{i=1}^n x_i\right) = \prod_{i=1}^n E(x_i)$$

It must be emphasised that the expectations of the product of a group of random variables are NOT ALWAYS equal to the product of their individual expectations. If the random variables are independent, then they are equal.

Example 60

Suppose that X_1 , X_2 , and X_3 are independent random variables such $E(X_i) = 0$ and $E(X_i^2) = 1$ for $i = 1, 2, 3$, Determine $E[(X_2 - 4X_3)^2]$

Solution

Since X_1 , X_2 and X_3 are independent, it follows that the two random variables X_1^2 and $(X_2 - 4X_3)^2$ are also independent. Therefore

$$\begin{aligned} E[X_1^2(X_2 - 4X_3)^2] &= E(X_1^2) E[(X_2 - 4X_3)^2] \\ &= E(X_1^2) E(X_2^2 - 8X_2X_3 + 16X_3^2) \\ &= E(X_1^2) - 8E(X_2X_3) + 16E(X_3^2) \text{ Since } E(X_i^2) = 1 \end{aligned}$$

$$= 1 - 8E(X_2)E(X_3) + 16$$

$$= 1 - 0 + 16 = 17$$

5.4 Expectation of Joint Probability Function

Let X and Y be random variables with joint probability distribution $f(x, y)$. The mean or expected value of the random variable $g(X, Y)$ is

$$\mu_{g(X, Y)} = E[g(X, Y)] = \sum \sum g(x, y) f(x, y) \quad \text{if } X \text{ and } Y \text{ are discrete, and}$$

$$\mu_{g(X, Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy \quad \text{if } X \text{ and } Y \text{ are continuous.}$$

5.3.1 Marginal Distribution

The marginal distribution of X alone and of Y alone are given by:

$$g(x) = \sum_y f(x, y); \quad h(y) = \sum_x f(x, y) \quad \text{for discrete variables}$$

$$g(x) = \int_y f(x, y) dy; \quad h(y) = \int_x f(x, y) dx \quad \text{for continuous variables}$$

5.3.2 Conditional Distribution

Let X and Y be two random variables, discrete or continuous. The ‘Conditional distribution’ of the random variable X , given that $Y = y$ is given by:

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h > 0$$

For the joint density function:

$$P(a < X < b | Y = y) =$$

$$\sum_x f(x | y) \quad \text{for discrete variable}$$

=

$$\int_a^b f(x | y) dx \quad \text{for continuous variable}$$

Example 61

The joint density for the random variable (x,y) is given by:

$$f(x, y) = \begin{cases} 10xy^2 & 0 < x < y < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Find the marginal densities g(x) and h(y) and the conditional density f(y|x).

Solution

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy = \frac{10xy^3}{3} \Big|_{y=x}^1 = \frac{10}{3}x(1 - x^3)$$

for $0 < x < 1$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = 5x^2y^2 \Big|_0^y = 5y^4 \quad \text{for } 0 < y < 1$$

Now,

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1 - x^3)} = \frac{3y^2}{(1 - x^3)}, \quad 0 < x < y < 1$$

Example 62

Let X and Y be the random variables with joint probability distribution given by the table below. Find the expected value of $g(X, Y) = XY$

		X		
		0	1	2
Y	f(x,y)			Row total
	0		$\frac{9}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	$\frac{3}{7}$
	2	$\frac{1}{28}$		$\frac{1}{28}$
Column total		$\frac{5}{14}$	$\frac{15}{28}$	1

Solution

(a)

$$\begin{aligned}
E(X, Y) &= \sum_{x=0}^2 \sum_{y=0}^2 xyf(x, y) \\
&= (0)(0)f(0, 0) + (0)(1)f(0, 1) + (0)(2)f(0, 2) \\
&\quad + (1)(0)f(1, 0) + (1)(1)f(1, 1) + (1)(2)f(1, 2) \\
&\quad + (2)(0)f(2, 0) + (2)(1)f(2, 1) + (2)(2)f(2, 2) \\
&= f(1, 1) = \frac{3}{14}
\end{aligned}$$

Example 63

Find

$$E\left(\frac{Y}{X}\right)$$

for the density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Solution

$$\begin{aligned}
E\left(\frac{Y}{X}\right) &= \int_0^1 \int_0^2 \frac{y(1+3y^2)}{4} dx dy = \frac{1}{4} [yx + 3y^3x]_0^2 dy \\
&= \frac{1}{4} \int_{y=0}^1 [2y + 6y^3] dy \\
&= \frac{1}{4} \left[y^2 + \frac{6}{4}y^4 \right]_0^1 = \frac{1}{4} \left[1 + \frac{6}{4} \right] = \frac{1}{4} \times \frac{10}{4} = \frac{5}{8}
\end{aligned}$$

5.5 Variance and covariance

The expected value of a random variable X is of special importance in statistics because it describes where the probability distribution is centred. By itself, however, the mean does not give adequate description of the shape of the distribution. We need to characterise the variability in the distribution.

5.4.1 Variance

Suppose that X is a random variable with mean, $\mu = E(X)$. The variance of X denoted by $\text{var}(X)$ is:

$$\text{Var}(X) = E[(X - \mu)^2] \text{ also } \text{Var}(X) = E(X^2) - \mu^2$$

It must be kept in mind that this may or may not be finite. If the expectation is not finite, it is said that $\text{Var}(X)$ does not exist. However, if the possible values of X are bounded, the $\text{Var}(X)$ must exist.

For a random variable with probability distribution $f(x)$ and mean μ , the variance of X is

$$\sigma^2 = \text{Var}(X) = E[(x - \mu)^2] = \sum (x - \mu)^2 f(x)$$

if X is discrete, and

$$\sigma^2 = \text{Var}(x) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

If X is continuous

Example 64

Let the random variable X represent the number of automobiles that are used for official business purposes on any given workday. The probability distributions for company A is given by:

X	1	2	3
f(x)	0.3	0.4	0.3

And for company B by:

X	0	1	2	3	4
f(x)	0.2	0.1	0.3	0.3	0.1

Show that the variances of the probability distribution for company B is greater than that of company A.

Solution

For company A

$$\mu = E(x) = \sum xf(x)$$

$$= (1)(0.3) + (2)(0.4) + (3)(0.3) = 2.0$$

and then

$$\sigma_A^2 = \sum_{x=1}^3 (x - \mu)^2 f(x) = \sum (x - 2)^2 f(x)$$

$$= (1 - 2)^2(0.3) + (2 - 2)^2(0.4) + (3 - 2)^2(0.3) = 0.6$$

For company B

$$\mu = E(x) = (0)(0.2) + (1)(0.1) + (2)(0.3) + (3)(0.3) + (4)(0.1) = 2.0$$

$$\sigma_B^2 = \sum_{x=0}^4 (x - 2)^2 f(x)$$

$$= (0 - 2)^2(0.2) + (1 - 2)^2(0.1) + (2 - 2)^2(0.3) + (3 - 2)^2(0.3) + (4 - 2)^2(0.1) = 1.6$$

$$\therefore \sigma_B^2 > \sigma_A^2$$

Properties of Variance

1. For any constants a and b :

$$\text{Var}(ax + b) = a^2 \text{Var}(x)$$

- (2) For any random variable X :

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

5.4.2 Covariance

When we consider the joint distribution of two random variables, the mean, the median and the variance provide useful information about their marginal distribution. However, these values do not provide any information about the relationship between the two variables or about their tendency to vary together rather than independently. The quantity covariance enables us to measure the association between two random variables, to determine the variance of the sum of any number of dependent random variables and to predict the value of one random variable by using the observed value of some other related variable.

Let X and Y be random variables having a specified joint distribution and Let $E(X) = \mu_x$, $E(Y) = \mu_y$, $\text{Var}(X) = \sigma_x^2$ and $\text{Var}(Y) = \sigma_y^2$. The covariance of X and Y , (denoted by $\text{COV}(X, Y)$), is defined as:

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_{xy} = E[(X - \mu_x)(Y - \mu_y)] \\ &= \sum \sum (X - \mu_x)(Y - \mu_y) f(x, y) \text{ if } X \text{ and } Y \text{ are discrete, and} \end{aligned}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y) f(x, y) dx dy \quad \text{if } X \text{ and } Y \text{ are continuous.}$$

It can be shown (DeGroot, 1989) that if $\sigma_x^2 < \infty$ and $\sigma_y^2 < \infty$, then the expectation above will exist and $\text{cov}(x, y)$ can be finite. However, the value of $\text{cov}(x, y)$ can be positive, negative or zero. When X and Y are independent covariance is zero. The converse however, is not generally true. Two variables may have zero covariance and still not be statistically independent. To dependent variables can be unrelated.

Some properties of covariance

1. For any random variables X and Y such

That $\sigma_x^2 < \infty$ and $\sigma_y^2 < \infty$,

$$\text{Cov}(X - Y) = E(XY) - E(X)E(Y).$$

Example 65

Two variables X and Y have a joint distribution as:

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{Elsewhere} \end{cases}$$

Find the covariance of X and Y

Solution

$$\sigma_{x,y} = E(X, Y) - \mu_x \mu_y$$

But $\mu_x = E(X)$ and $\mu_y = E(Y)$

Hence we first must compute the marginal density functions But for a joint distribution $f(x, y)$ marginal distributions of X and Y alone are given by:

$$g(x) = \sum_y f(x, y) \text{ and } h(y) = \sum_x f(x, y) \text{ for discrete variable}$$

$$g(x) = \int_y f(x, y) dy \text{ and } h(y) = \int_x f(x, y) dx \text{ for continuous variable}$$

Thus:

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5} (2x + 3y) dy \\ &= \frac{4xy}{5} + \frac{6y^2}{10} \Big|_0^1 = \frac{4x + 3}{5} \end{aligned}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5} (2x + 3y) dx = \frac{2(1 + 3y)}{5}$$

$$E(X) = \int_0^1 \frac{x(4x + 3)}{5} dx = \frac{1}{5} \int_0^1 4x^2 + 3x = \frac{1}{5} \left(\frac{4}{3}x^3 + \frac{3}{2}x^2 \right) \Big|_0^1$$

=

$$\frac{1}{5} \left(\frac{4}{3} + \frac{3}{2} \right) = \frac{17}{30} \quad \dots\dots (1)$$

It could have been solved from

$$\int_{x=0}^1 \int_{y=0}^1 xf(xy) dx dy$$

$$E(y) = \int \frac{2}{5} (1 + 3y) dy = \frac{2}{5} \int_0^1 (y + 3y^2) dy$$

$$\begin{aligned} &= \frac{2}{5} \left(\frac{1}{2}y^2 + y^3 \right) \Big|_0^1 \\ &= \frac{2}{5} \left(\frac{1}{2} + 1 \right) = \frac{2}{5} \times \frac{3}{2} = \frac{3}{5} \quad \dots\dots (2) \end{aligned}$$

It also could have been solved from

$$\int_{x=0}^1 \int_{y=0}^1 yf(xy) dx dy$$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^1 xy \frac{2}{5} (2x + 3y) dy dx \\ &= \frac{2}{5} \int_{y=0}^1 \int_{x=0}^1 (2x^2y + 3xy^2) dx dy \end{aligned}$$

$$\begin{aligned} &= \frac{2}{5} \int \left(\frac{2}{3}x^3y + \frac{3}{2}x^2y^2 \right) \Big|_0^1 dy \\ &= \frac{2}{5} \int \frac{2}{3}y + \frac{3}{2}y^2 dy \end{aligned}$$

$$\begin{aligned} &= \frac{2}{5} \left(\frac{1}{3}y^2 + \frac{1}{2}y^3 \right) \Big|_0^1 \\ &= \frac{2}{5} \left(\frac{1}{3} + \frac{1}{2} \right) = \left(\frac{2}{5} \times \frac{5}{6} \right) = \frac{1}{3} \quad \dots\dots (3) \end{aligned}$$

$$\begin{aligned}\therefore \sigma_{xy} &= E(XY) - E(X)E(Y) \\ &= (3) - (1)(2) \\ &= \frac{1}{3} - \left(\frac{17}{30} \times \frac{3}{5}\right) \\ &= 0.333 - 0.34 \\ &\approx -0.01\end{aligned}$$